Homework 6
Statistics 2103

Please submit your code, as well as output (including plots), for all questions.

1. Suppose we want to study the association between a car's weight and its fuel efficiency. The file `mileage.csv` contains the following data for 398 cars:

   `mpg`: fuel efficiency, in miles per gallon
   `weight`: vehicle weight in pounds

   (a) Fit a linear regression model, using `weight` as the predictor and `mpg` as the response. Report the estimated model coefficients.

   (b) According to the model in part (a), what is the predicted fuel efficiency for a 5000 pound car?

   (c) Make a scatter plot of the residuals from the model in part (a), with `weight` on the x axis. Based on this plot, do any of the linear regression assumptions appear to be violated?

   (d) Fit a new linear regression model, using log(mpg) as the response (log refers to natural log). Report the model coefficients.

   (e) Describe the interpretation of the slope coefficient in this model.

   (f) Make a scatter plot of the residuals from the model in part (d), with `weight` on the x axis. Based on this plot, do any of the linear regression assumptions appear to be violated? Does this look like an improvement over the original model?

   (g) According to the model in part (d), what is the predicted fuel efficiency for a 5000 pound car?

2. The file `voting.csv` contains voting data from a random sample of 70 voting stations in a particular region in a past election. The variables in the data are:

   `turnout`: the percentage of registered voters at that voting station who voted in the election
   `dem_votes`: the percentage of total votes cast at that voting station that were for the Democratic party candidate

(a) Fit a linear regression model, using `turnout` as the predictor and `dem_votes` as the response. Report the estimated model coefficients, and display the fitted regression line on a scatter plot of the data.

(b) There are two outliers in the data set (points inconsistent with the pattern in the rest of the data). One of them has an unusual y value, and the other has an unusual x value. What are the (x,y) values of those two data points?

Tip: If you are struggling with this, try printing the first few rows of the data set.

(c) Can you justify removing the outliers identified in part (b) from the data set? Why or why not?

(d) Regardless of your answer to part (c), remove the data point with the unusual y value. Fit a new linear regression model to the rest of the data, and report the new slope.

(e) Regardless of your answer to part (c), remove the data point with the unusual x value (but **don't** remove the other outlier). Fit a new linear regression model to the rest of the data, and report the new slope.

(f) Make a scatter plot of the full data set, and display all three regression lines (from parts (a), (d), and (e)) on the plot.

(g) Based on your results in parts (d)-(f), which of the two outliers appears to have more influence on the slope of the least squares regression line?

3. We are interested in predicting the risk that patients at a hospital acquire infections during their stay, using the characteristics of the hospital and its patient population. The file `SENIC.csv` contains the following variables:

`IR` (Y): infection risk - percentage of patients acquiring infection in hospital (in percentage points, i.e. values range from 0 to 100)
`Age` (X_1): average age of patients
`Stay` (X_2): average length of stay, in days
`Beds` (X_3): number of beds in hospital
`Services` (X_4): percentage of 35 particular services available at the hospital (values range from 0 to 100)
`Nurses` (X_5): number of full-time equivalent nurses at hospital

(a) Make a histogram of the response variable, infection risk (IR).

(b) Fit a linear regression model to predict infection risk using only number of beds (X_3)

as a predictor. State the slope and intercept, and describe the interpretation of the slope coefficient.

(c) Fit a linear regression model to predict infection risk using all five predictors. What is the coefficient for number of beds (X_3) in this model? Describe the interpretation of that value.

(d) Use the models from part (b) and part (c) to predict the infection risk for patients at a hospital with the following characteristics:

(Stay = 7, Age = 45, Beds = 550, Nurses = 115, Services = 40)

Compare the two predictions.

(e) In the multiple linear regression model from part (c), perform a test of the hypothesis that the coefficient for `Age` (X_1) equals zero (using a 5% rejection threshold). State your conclusion, and describe the meaning of that conclusion in context.

(f) Is the linear model from part (c) useful overall? State the relevant null and alternative hypotheses for this question. (We will cover this topic in class on Wednesday November 4.)

(g) Perform the hypothesis test from part (f). Report the p-value and state your conclusion (using a 5% rejection threshold), including an answer to the original question. (We will cover this topic in class on Wednesday November 4.)